Percentages

1 What does the $\%$ symbol mean when placed after a number?

2 Fill in the blanks to make the following statements true:

a 29% is equivalent to 29 out of $\square$.

b 53% is equivalent to $\square$ out of 100.

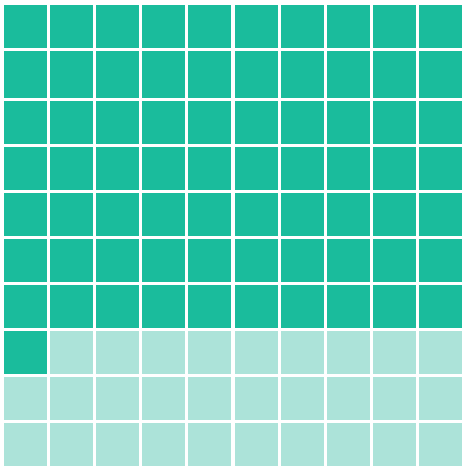
3 Given the grids below, answer the following:

i How many squares are shaded?

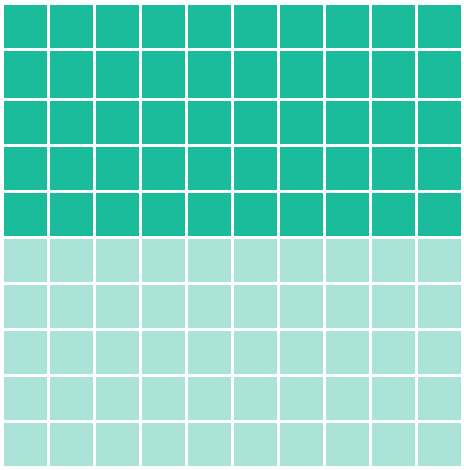
ii What percentage of the grid is shaded?

iii What fraction of the grid does this percentage represent?

a



b



Converting from percentages

4 Write the following as fractions in their simplest form:

a 68% **b** 13% **c** 55% **d** 275%

e 37% **f** 94% **g** 450% **h** 2.5%

5 Write the following as decimals:

a 54% **b** 60% **c** 270% **d** 303%

e 76% **f** 500% **g** 182% **h** 6897%

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Converting to percentages

6 For each of the following fractions:

i Write an equivalent fraction with a denominator of 100.

ii Write it as a percentage.

a $\frac{5}{10}$ b $\frac{9}{300}$ c $\frac{1}{8}$ d $\frac{45}{200}$

7 Write the following as percentages:

a $\frac{83}{100}$ b $1\frac{4}{5}$ c $\frac{42}{100}$ d $\frac{39}{60}$

e $\frac{93}{150}$ f $\frac{21}{35}$ g $3\frac{7}{10}$ h $2\frac{9}{50}$

8 Consider the fraction $\frac{2}{3}$.

a Is there an equivalent fraction that has a denominator of 100, and a whole number as its numerator?

b What is the closest whole percent to $\frac{2}{3}$?

9 For each of the following pairs, either identify that they are the same or state which number is larger:

a $\frac{1}{8}$, 13% b $\frac{4}{25}$, 16% c $\frac{63}{90}$, 70% d $\frac{1}{3}$, 27%

10 Write the following as a percentages:

a 0.24 b 1.1 c 0.857 d 8

e 1.71 f 24 g 13.04 h 0.041

Comparing different values

11 When converting a fraction or decimal to a percentage, why is it necessary to multiply by 100%?

12 When converting from percentage to a fraction or decimal, why is it necessary to divide by 100%?

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13 Convert between percentages, fractions and decimals to complete the table:

Fraction	Decimal	Percentage
$\frac{11}{100}$		
	1.83	
$\frac{5}{8}$		

14 Express the following as both a fraction and a decimal:

a 80% **b** $47\frac{1}{2}\%$

15 For the following groups of numbers:

i Rewrite the group as percentages

ii Hence, order the original group from smallest to largest.

a $\frac{5}{10}, \quad 100\%, \quad 0.3$ **b** $\frac{1}{4}, \quad 60\%, \quad 0.3$ **c** $\frac{3}{8}, \quad 20\%, \quad 0.8$ **d** $\frac{3}{15}, \quad 45\%, \quad 0.27$

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